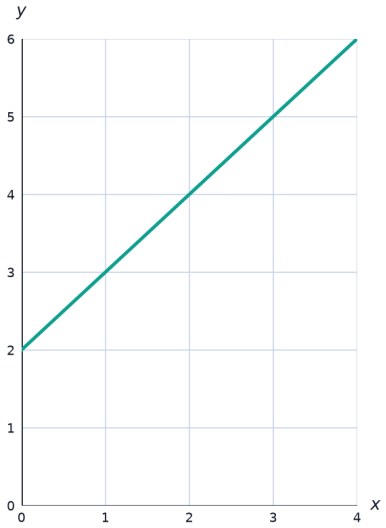
Pre-Calculus Concepts

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

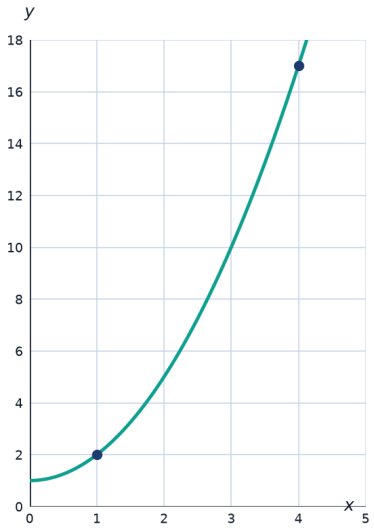
1 **[Foundational]** • **Target: Limits: an introductory numerical approach**

Estimate $\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$ by first simplifying the expression algebraically.



2 **[Foundational]** • **Target: Average rate of change**

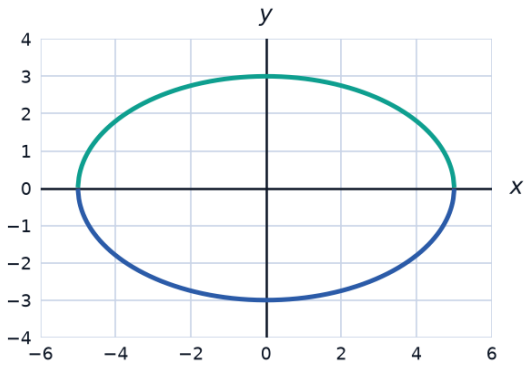
Find the average rate of change of $f(x) = x^2 + 1$ over the interval $[1, 4]$.



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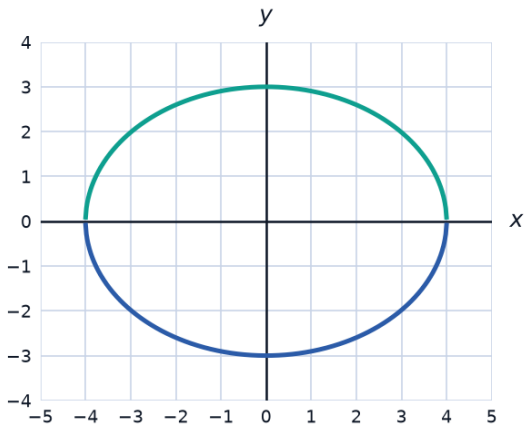
3 [Foundational] • Target: Conic sections: ellipses

Find the equation of an ellipse centered at the origin with $a = 5$ (horizontal semi-axis) and $b = 3$ (vertical semi-axis).



4 [Foundational] • Target: Conic sections: ellipses

For the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$, find the coordinates of the vertices along the major axis.



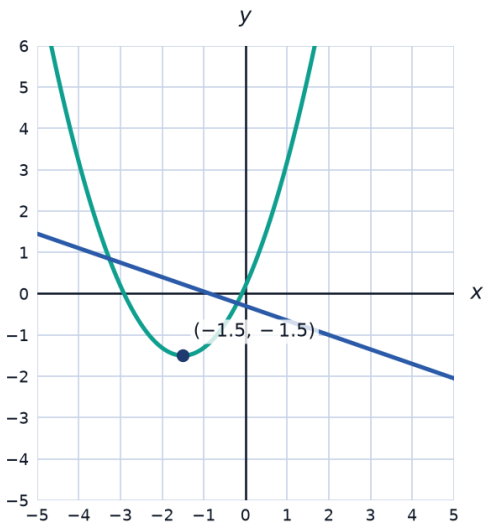
5 [Foundational] • Target: Conic sections: hyperbolas

Find the equation of a hyperbola centered at the origin with vertices at $(\pm 4, 0)$ and $b = 3$.

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6 [Foundational] • Target: Parametric equations

Eliminate the parameter from $x = t + 1$ and $y = t^2$, expressing y in terms of x.



7 [Foundational] • Target: Sequences and series: sigma notation

Write $4 + 8 + 12 + 16 + 20$ using sigma notation, and evaluate it.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Binomial theorem

Use the binomial theorem to expand $(x + 2)^4$.

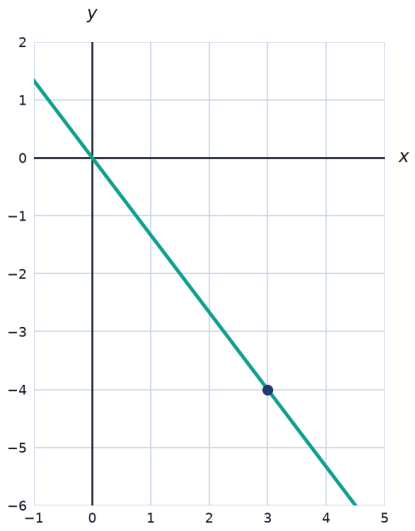
9 [Intermediate] • Target: Binomial theorem

Find the coefficient of x^3 in the expansion of $(x + 3)^5$, using $\binom{5}{k}$.

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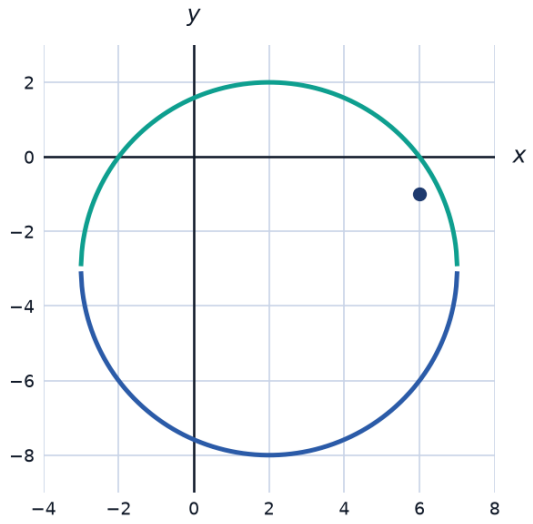
10 [Intermediate] • Target: Vectors in the plane

Given $\mathbf{u} = (3, -4)$, find its magnitude and a unit vector in the same direction.



11 [Intermediate] • Target: A multi-part conic and function analysis problem

This question has two parts. A circle has equation $x^2 + y^2 - 4x + 6y - 12 = 0$. (a) Complete the square to write the equation in standard form, and state the center and radius. (b) Determine whether the point $(6, -1)$ lies inside, on, or outside the circle.



12 [Intermediate] • Target: Matrix operations

Matrix A has rows $(2, -1)$ and $(3, 4)$; matrix B has rows $(0, 5)$ and $(-2, 1)$. Find $A - B$.

13 [Intermediate] • Target: Infinite geometric series

Find the sum of the infinite geometric series $8 + 4 + 2 + 1 + \cdots$, using $S = \frac{a_1}{1 - r}$.

14 [Intermediate] • Target: Infinite geometric series

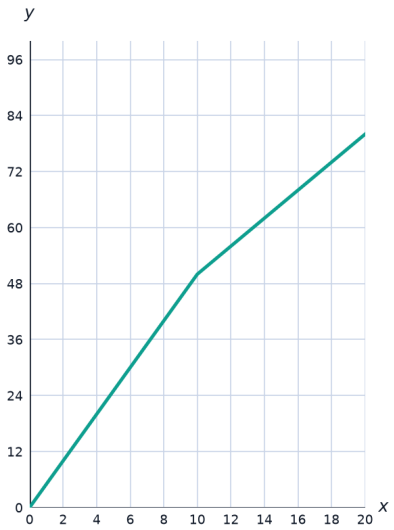
Write the repeating decimal $0.\overline{6}$ as a fraction, by treating it as an infinite geometric series $\frac{6}{10} + \frac{6}{100} + \frac{6}{1000} + \cdots$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging] • Target: Piecewise and step functions in applied contexts • Cross-Lesson: 6.01 Advanced Coordinate Geometry**

A shipping company charges 5 dollars per pound for the first 10 pounds, then 3 dollars per pound for each pound beyond 10. Write a piecewise function $C(w)$ for the cost, and find the cost of a 15-pound package.

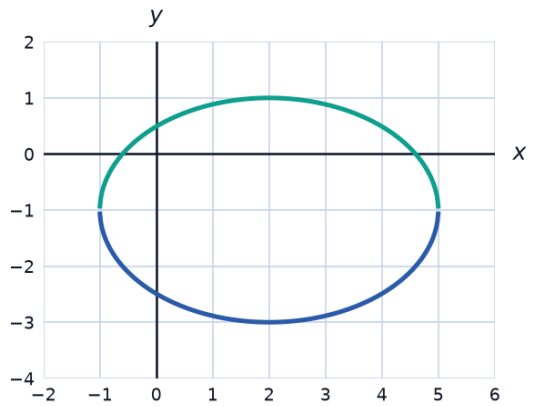


16 **[Challenging] • Target: Finding terms of a sequence defined recursively • Cross-Lesson: 7.01 Advanced Plane Geometry**

A sequence is defined by $a_1 = 3$ and $a_n = 2a_{n-1} + 1$ for $n \geq 2$. Find a_4.

17 **[Challenging] • Target: A multi-part conic section identification and analysis problem • Cross-Lesson: 8.01 Statistics, Counting & Probability**

This question has three parts. An equation is given: $4x^2 + 9y^2 - 16x + 18y - 11 = 0$. (a) Complete the square in both x and y to write the equation in standard conic form. (b) Identify the type of conic section. (c) State the center and the lengths of the semi-axes.



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18 [Challenging] • Target: Conic sections: hyperbolas • Cross-Lesson: 1.01 Advanced Number Theory

Find the 4th term in the binomial expansion of $(2x - 3)^6$.

19 [Challenging] • Target: Parametric equations • Cross-Lesson: 2.01 Advanced Algebra & Systems

Given vectors $\vec{u} = \langle 3, -4, 2 \rangle$ and $\vec{v} = \langle 2, 1, k \rangle$, find the value of k such that $\vec{u}$ and $\vec{v}$ are orthogonal.

20 [Challenging] • Target: Sequences and series: sigma notation • Cross-Lesson: 3.01 Advanced Functions

An infinite geometric series has common ratio $r = -0.4$ and converges to an infinite sum of $S_\infty = 25$. Find the first term a_1, and compute the exact value of the partial sum S_4.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Limits: an introductory numerical approach

For the ellipse $\frac{x^2}{1681} + \frac{y^2}{81} = 1$ (horizontal major axis), find the coordinates of the foci.

a. $(\pm 9, 0)$

b. $(0, \pm 40)$

c. $(\pm 40, 0)$

d. $(\pm 41, 0)$

22 [Foundational] • Target: Average rate of change

A hyperbola has vertices at $(\pm 20, 0)$ and $b = 21$. Find the coordinates of the foci.

a. $(\pm 29, 0)$

b. $(\pm 20, 0)$

c. $(\pm 21, 0)$

d. $(\pm 41, 0)$

23 [Foundational] • Target: Conic sections: ellipses

Find the sum of the infinite geometric series with first term $a_1 = 12$ and common ratio $r = \frac{1}{3}$, using $S = \frac{a_1}{1 - r}$.

a. 9

b. 36

c. 18

d. 8

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24 [Foundational] • Target: Conic sections: hyperbolas

In the standard (x, y) coordinate plane, what is the equation of the horizontal asymptote of the graph of the function $f(x) = \frac{8x+3}{2x-5}$?

a. $y = 4$

b. $y = \frac{5}{2}$

c. $x = \frac{5}{2}$

d. $y = -\frac{3}{5}$

25 [Foundational] • Target: Parametric equations

A geometric sequence has its first term $a_1 = 5$ and its common ratio $r = 2$. If the sum of the first n terms of this sequence is $S_n = 315$, find the value of n.

a. 4

b. 5

c. 6

d. 7

26 [Foundational] • Target: Sequences and series: sigma notation

What is the maximum value attained by the function $f(x) = \frac{\sin x - 2\cos x + 4}{2\sin x + \cos x + 3}$ for all real numbers x?

a. 3

b. 4

c. $\frac{9}{2}$

d. $\frac{11}{2}$

27 [Foundational] • Target: Binomial theorem

In the standard (x, y) coordinate plane, what is the equation of the vertical asymptote of the graph of $f(x) = \frac{3x+1}{x-4}$?

a. $y = 3$

b. $x = 4$

c. $x = -\frac{1}{3}$

d. $y = 4$

28 [Foundational] • Target: Vectors in the plane

What is the period of the function $f(x) = 3\cos(2x)$?

a. $\frac{\pi}{2}$

b. π

c. 2π

d. 4π

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: A multi-part conic and function analysis problem

A geometric sequence has first term $a_1 = 3$ and common ratio $r = 3$. Find a_4.

a. 9

b. -81

c. 243

d. 81

30 [Intermediate] • Target: Matrix operations

A sequence is defined by $a_1 = 1$ and $a_{n+1} = a_n + 2n + 1$ for $n \geq 1$. Prove by induction that $a_n = n^2$ for all positive integers n. Then, consider the series $S_N = \sum_{n=1}^{N} a_n$. Find a closed-form expression for S_N and use it to determine the value of $\sum_{n=1}^{10} (a_n - a_{n-1})$ where $a_0 = 0$.

a. 100

b. 110

c. 10

d. 210

31 [Intermediate] • Target: Infinite geometric series

A sequence is defined by the recurrence relation $a_{n+1} = a_n^2 - 2$ for $n \geq 1$. If $a_1 = 2$, find the value of a_{2023}.

a. 2

b. 0

c. -2

d. 1

32 [Intermediate] • Target: Piecewise and step functions in applied contexts

Find S_∞ for the geometric series with first term -8 and common ratio $\frac{1}{4}$.

a. $\frac{32}{3}$

b. $-\frac{32}{3}$

c. -6

d. $-\frac{32}{5}$

33 [Intermediate] • Target: Finding terms of a sequence defined recursively

A geometric series has $a_1 = 3$ and $r = 2$. If $S_n = 21$, find n.

a. 5

b. 1

c. 4

d. 3

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34 **[Intermediate]** • **Target: A multi-part conic section identification and analysis problem**

A function is defined by $f(x) = 2x - 4$ for all x near $x = -4$ (including at $x = -4$). What is $\lim_{x \to -4} f(x)$?

a. -15

b. -12

c. -14

d. -10

35 **[Intermediate]** • **Target: Limits: an introductory numerical approach**

The sequence $-7, -10, -13, -16, \ldots$ continues with the same pattern. Find the 14th term.

a. -43

b. -46

c. 46

d. -50

36 **[Intermediate]** • **Target: Average rate of change**

An arithmetic sequence (a_n) and a geometric sequence (b_n) of non-zero real numbers share the same first term, so $a_1 = b_1$. The sequences also satisfy $a_5 = b_2$ and $a_6 = b_3$, with $b_2 eq b_1$. If the maximum value of the partial sum $S_N = \sum_{k=1}^{N} a_k$ of the arithmetic sequence is 153, what is the sum of the infinite geometric series $\sum_{n=1}^{\infty} b_n$?

a. 64

b. 96

c. 144

d. 192

37 **[Intermediate]** • **Target: Conic sections: ellipses**

An infinite geometric sequence of positive real numbers has first term a_1 and common ratio $r \in (0, 1)$. Let $S_1 = \sum_{n=1}^{\infty} a_n$, $S_2 = \sum_{n=1}^{\infty} a_n^2$, and $S_3 = \sum_{n=1}^{\infty} a_n^3$. If $S_1 = 4$ and $S_2 = 8$, what is the value of S_3?

a. $\frac{256}{13}$

b. $\frac{128}{13}$

c. $\frac{64}{7}$

d. 16

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Conic sections: hyperbolas • Cross-Lesson: 1.01 Advanced Number Theory

The first term of a geometric sequence is a. The common ratio is r. The sum of the first 3 terms is 21, and the sum of the first 6 terms is 189. If $r > 0$, find the value of a and r.

a. $a = 3, r = 2$

b. $a = 7, r = 1/2$

c. $a = 3, r = \sqrt{2}$

d. $a = 21, r = 1$

39 [Challenging] • Target: Parametric equations • Cross-Lesson: 2.01 Advanced Algebra & Systems

The first three terms of a geometric sequence of positive real numbers are $k - 1$, $k + 2$, and $3k$, in that order. What is the value of the 6th term of this sequence?

a. 24

b. 48

c. 96

d. 192

40 [Challenging] • Target: Sequences and series: sigma notation • Cross-Lesson: 3.01 Advanced Functions

The second term of a geometric sequence of positive real numbers is k, and the fifth term is $k^2 - 20$. If the common ratio of the sequence is 2, what is the sum of the first 6 terms of the sequence?

a. 155

b. 160

c. 315

d. 630

41 [Challenging] • Target: Binomial theorem • Cross-Lesson: 4.01 Advanced Trigonometry

Consider the arithmetic sequence with first term $a_1 = 5$ and a common difference $d = 3$. What is the formula for the n-th term, a_n?

a. $a_n = 5 + 3(n - 1)$

b. $a_n = 3n + 2$

c. $a_n = 5n + 3$

d. $a_n = 5 + 3n$

42 [Challenging] • Target: Vectors in the plane • Cross-Lesson: 6.01 Advanced Coordinate Geometry

A geometric sequence has its first term a_1 and common ratio r. The sum of the first 3 terms is 21, and the sum of the first 6 terms is 189. Find the value of a_1 given that $req1$.

a. 1

b. 3

c. 7

d. 9

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43 [Challenging] • Target: A multi-part conic and function analysis problem • Cross-Lesson: 7.01 Advanced Plane Geometry

In the standard (x, y) coordinate plane, the graph of $y = f(x)$ contains the point $(-2, 5)$. The function g is defined by $g(x) = f(x - 3) - 4$. Which of the following points must lie on the graph of $y = g(x)$?

a. $(1, 1)$

b. $(-5, 1)$

c. $(1, 9)$

d. $(-5, 9)$

44 [Challenging] • Target: Matrix operations • Cross-Lesson: 8.01 Statistics, Counting & Probability

In a geometric sequence of positive real numbers, the sum of the first two terms is 18, and the sum of the first four terms is 90. What is the fifth term of the sequence?

a. 48

b. 72

c. 96

d. 192

45 [Challenging] • Target: Infinite geometric series • Cross-Lesson: 1.01 Advanced Number Theory

Consider an arithmetic sequence where the 5th term is 19 and the 11th term is 43. If the first term is denoted by a_1 and the common difference by d, find the value of $a_1 + d$.

a. 6

b. 7

c. 8

d. 9

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